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Building the Next Cybersecurity Workforce: A Grades 7–12 Curriculum to Close the Cyber Talent Gap

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Abstract

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Keywords

Digital Forensics; Cybersecurity Curriculum; Cybersecurity Workforce; Cyber Talent Gap; STEM Education; Cybersecurity Awareness

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Building the Next Cybersecurity Workforce: A Grades 7–12 Curriculum to Close the Cyber Talent Gap

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***Abstract*— In today's rapidly evolving technological landscape, cyberattacks pose increasing threats, yet a global shortage of cybersecurity and digital forensics professionals leaves industries vulnerable, similar to having too few law enforcement officers in a densely populated city. The judicial system faces rising digital crimes and fraud cases, further strained by the lack of experts to analyze and extract digital evidence. Despite high demand, millions of positions remain unfilled. This paper identifies the root causes of the cybersecurity workforce shortage and proposes a targeted solution: a curriculum for Grades 7–12 designed to foster cybersecurity awareness and interest. The methodology included a comprehensive literature review, expert consultations, curriculum mapping, and a pilot survey of 137 students across four private schools. Survey results revealed that 77% of students were aware of cybersecurity concepts, 89% recognized its future importance, and 67% advocated for hands-on cybersecurity labs in schools. This adaptable program can be integrated into educational institutions to equip students with foundational skills and encourage early career exploration in cybersecurity. The initiative addresses the skills gap and aims to cultivate a workforce prepared for the future.**

Keywords Digital Forensics; Cybersecurity Curriculum; Cybersecurity Workforce; Cyber Talent Gap; STEM Education; Cybersecurity Awareness

I. Introduction

Cybersecurity threats continue to escalate across sectors and geographies, intensifying risk for enterprises, critical infrastructure, and public institutions. Recent global outlooks emphasize that resource and skills gaps have become primary barriers to cyber resilience, with many organizations reporting material impacts from breaches and prolonged recovery timelines. These trends are compounded by emerging technologies (e.g., generative AI), legacy systems, and expanding attack surfaces, all of which raise the bar on workforce capabilities and capacity.

Against this backdrop, the demand for cybersecurity professionals has surged at an unprecedented rate, outpacing employers' ability to fill critical roles, a trend projected to persist. Cybersecurity Ventures reported a 350% increase in the global cybersecurity job gap from 1 to 3.5 million between 2013 and 2021, with forecasts suggesting a sustained shortage in the near term [1]. Complementing this historical view, CyberSeek's mapping of demand and supply has highlighted acute talent gaps across U.S. states and metro areas, offering granular workforce intelligence that helps stakeholders gauge local challenges and opportunities [2]. In the U.S., employers posted 714,548 cybersecurity job openings from April 2021 to April 2022, with postings up 43% year over year, far exceeding the ~18% growth across the broader labor market [3].

The most recent global statistics sharpen the picture. According to the 2023 ISC² Cybersecurity Workforce Study, the global workforce gap reached 4 million professionals (a 12% increase from 2022), even as the active workforce climbed to 5.5 million [4]. In Europe, ENISA reports persistent and structural shortages, estimated at hundreds of thousands of professionals, limiting the region's ability to protect critical sectors and comply with accelerating regulatory requirements [5]. In the United States, the Bureau of Labor Statistics (BLS) projects employment for information security analysts to grow 32% from 2022 to 2032, far outpacing the average for all occupations [6]. These developments reinforce that cybersecurity talent pipelines must expand rapidly to meet global and regional needs.

New data from ISC² (2024) suggests the gap widened further, with multiple analyses converging on an approximate shortfall of ~4.7–4.8 million professionals, as workforce growth slowed and budget constraints weighed on teams. These findings align with third-party syntheses that flag budget pressure, mid/advanced role hiring bias, and skills deficits (e.g., AI, cloud security, zero trust, DFIR) as pervasive contributors to the gap [7]. The World Economic Forum (2024) likewise highlights a near 4 million global shortage and calls for coordinated public–private action to close the skills gap [8].

While demand remains high globally, the workforce gap is not uniform. CyberSeek's latest dashboards present a global workforce estimate range of 4.4–5.5 million, with U.S. openings >500,000 in the most recent period, evidence that the market continues to expand and diversify across industries beyond traditional tech (e.g., finance, manufacturing, and public sector) [9]. Enterprises increasingly report that skills shortages correlate with breach frequency, higher recovery times, and seven-figure incident costs, underscoring the operational consequences of understaffed security teams [10].

Existing initiatives such as CyberPatriot (Air & Space Forces Association) and NICE (National Initiative for Cybersecurity Education) have made valuable contributions to K–12 awareness through competitions, cybercamps, and role exploration aligned to standardized frameworks. However, these efforts are rarely embedded as structured, multi-year curricula within formal education systems. To help bridge the workforce gap upstream, this paper proposes a comprehensive Grades 7–12 curriculum that provides early exposure, sustained engagement, and hands-on experiences across foundational cybersecurity domains, including digital forensics, secure networking, and applied defensive techniques. The curriculum maps onto recognized standards (e.g., NIST NICE Framework SP 800-181 Rev. 1 and competency areas) to support coherent progression, credential alignment, and pathways into apprenticeships and industry certifications [11], [12], [13], [14].

This approach complements broader policy and ecosystem advances. In the EU, the Cybersecurity Skills Academy (European Commission, 2023) aims to orchestrate training, funding, and measurement across stakeholders and to leverage the ENISA European Cybersecurity Skills Framework (ECSF) for consistent role definitions and capability assessment, an approach intended to boost competitiveness, resilience, and compliance readiness across Member States [11], [12]. In the U.S., the NICE Framework and the National K12 Cybersecurity Education Roadmap provide explicit guidance for curriculum development, career pathways, and multi-stakeholder engagement across K–12 and higher education [13], [14].

Section 2 of this manuscript reviews the current scientific and policy landscapes, with particular attention to root causes of the talent shortfall. Among the factors discussed, the absence of formal cybersecurity education in middle and high schools emerges as a primary driver. Despite ubiquitous technology use among adolescents, many lack foundational awareness of cybersecurity, data privacy, and digital forensics, and even motivated computer science students often have limited exposure to secure development practices. These findings echo workforce studies that consistently list cloud security, AI/ML, zero trust, and DFIR as top skills gaps [4], [7], [15].

Accordingly, the Grades 7–12 curriculum is designed to operationalize early learning with hands-on labs, competency-based assessments, and alignment to recognized work roles, thereby supporting students' transition to advanced study, apprenticeships, and entry-level cyber roles. The program's modular structure enables integration into diverse school contexts (public, private, urban, rural), with adaptability to regional workforce needs revealed by tools like the CyberSeek heat map [2], [9]. Rather than functioning as isolated course offerings, the curriculum is structured as a coherent, standards-aligned framework that schools can tailor to their scheduling models, infrastructure capacity, and state-level requirements while preserving defined learning outcomes and competency benchmarks.

To illustrate scale and trajectory, *Figure 1a* and *Figure 1b* visualize the global cybersecurity workforce gap and global workforce estimate (2022), contextualizing historical growth and signaling why pipeline interventions must begin earlier in the education continuum [15]. Together with the latest ISC², ENISA, BLS, and WEF data, these figures motivate the curriculum's design choices and evaluation methodology.

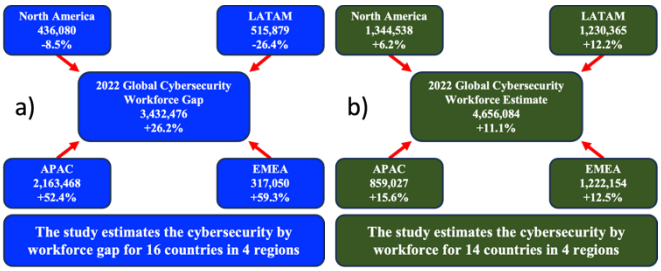


Fig. 1. Global Security Workforce (a) 2022 Global Cybersecurity Workforce Gap; (b) 2022 Global Cybersecurity Workforce Estimate

II. The Present State of Cyberattacks and the Scarcity of Qualified Cybersecurity Talent

A. Current Cyberattacks and Vulnerabilities

In the rapidly advancing digital landscape, society benefits from unprecedented connectivity and productivity; yet the same forces expand our exposure to online threats and exploitable weaknesses. Recent literature characterizes the cyber threat environment as continuously evolving, with impacts felt by large enterprises, government agencies, small businesses, and individuals alike [16]. This subsection synthesizes prevailing attack trends, representative incidents, and emerging vulnerabilities to frame the urgency of workforce development and education.

Cyberattacks Landscape and Types**:** As technology progresses, attack breadth and sophistication grow in tandem. Threat actors regularly adapt to defenses by combining social engineering with technical exploitation, targeting users, endpoints, applications, and cloud workloads [16]. *Figure 2* summarizes common tactics. Phishing remains the primary delivery mechanism for initial compromise, leveraging trusted branding and persuasive context to induce credential disclosure or malware execution. Ransomware persists at scale: adversaries penetrate networks, laterally move, exfiltrate data, encrypt critical assets, and coerce payment for decryption keys and non-publication of stolen data. Service disruption tactics, DoS/DDoS, intentionally overwhelm networked services to degrade availability, often used as standalone attacks or as diversions. Zero-day exploits weaponize previously unknown software flaws to gain unauthorized execution or escalate privileges before vendors can issue patches. Finally, spoofing and related impersonation techniques mislead users by mimicking legitimate domains, services, or communications [16].

Notable Cyberattacks: High-profile incidents underscore systemic risk across supply chains and critical infrastructure. The *SolarWinds* campaign (2020) compromised a widely deployed network management platform via tainted software updates, enabling follow-on intrusions in public and private networks and spotlighting dependencies in the software supply chain as shown in *Figure 3* [17], [18]. In 2021, the *Colonial Pipeline* ransomware incident disrupted fuel distribution across the U.S. East Coast, demonstrating how operational technology and business networks intersect in ways that amplify societal impact [17], [18]. Likewise in 2021, the *JBS* ransomware attack

temporarily halted meat processing operations, revealing vulnerabilities in global logistics and food supply chains [17], [18]. These exemplars reveal how diverse entry points, update channels, remote access, exposed services, and phishing, translate into outsized operational consequences.

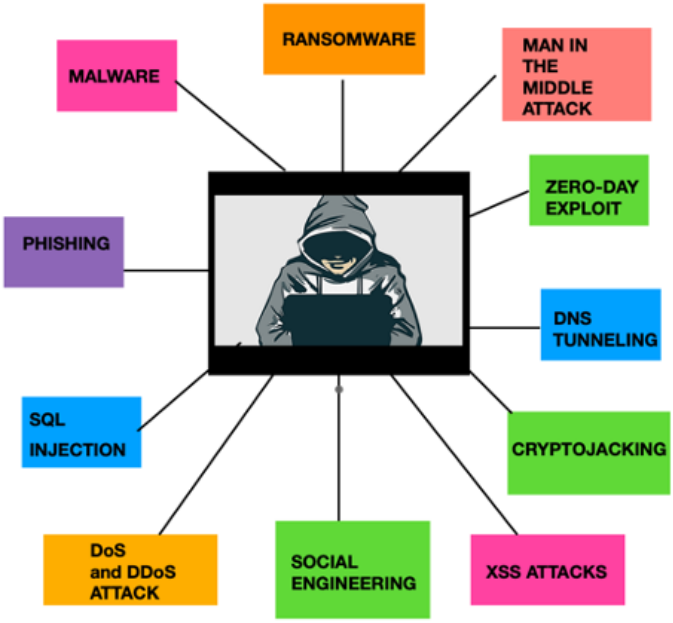


Fig. 2. Types of Cyberattacks

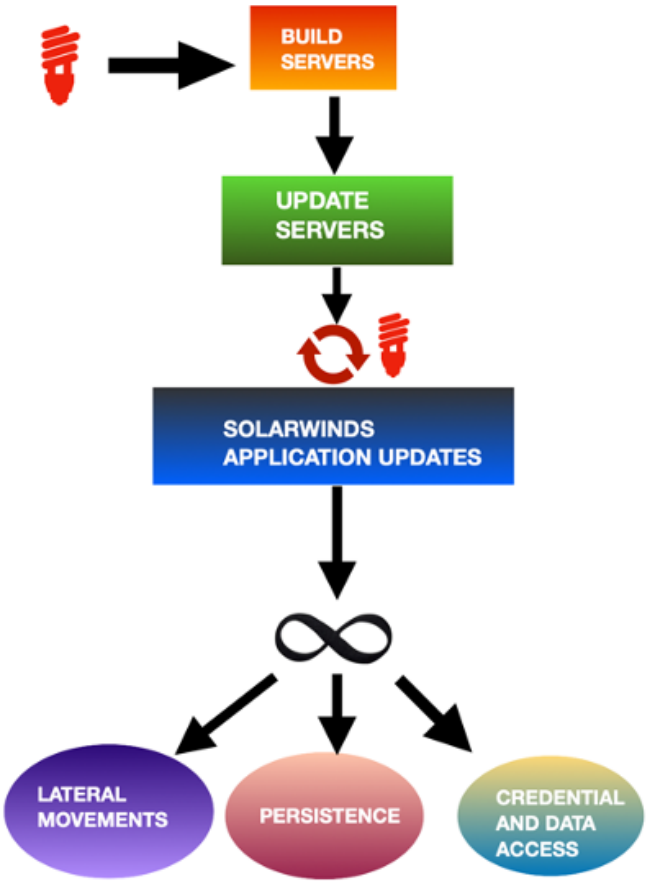


Fig. 3. Footprints of SolarWinds Cyberattack

Current Vulnerabilities: Adversaries commonly exploit human error (e.g., interacting with malicious links/attachments), unpatched software defects, weak or reused passwords, and insecure configurations that expose administrative interfaces or default settings. Routine, timely patching reduces risk, but coverage gaps persist across complex IT estates, especially where legacy systems and third-party dependencies are present [18], [19]. Together, these weaknesses reinforce the need for baseline cyber hygiene across users and administrators.

Emerging Vulnerabilities: Rapid adoption of IoT devices expands the attack surface through heterogeneous firmware, limited security controls, and inconsistent update mechanisms as shown in *Figure 4* [19], [20]. Cloud migration introduces configuration errors (e.g., overly permissive identities, exposed storage), necessitating shared-responsibility comprehension and secure-by-default orchestration. Meanwhile, both attackers and defenders leverage AI/ML: defenders accelerate detection and response, while adversaries use automation and synthetic content to scale spear-phishing, evasion, and data analysis [17], [18]. These developments intensify skill requirements for practitioners in identity, network segmentation, secure configuration, incident response, and digital forensics.

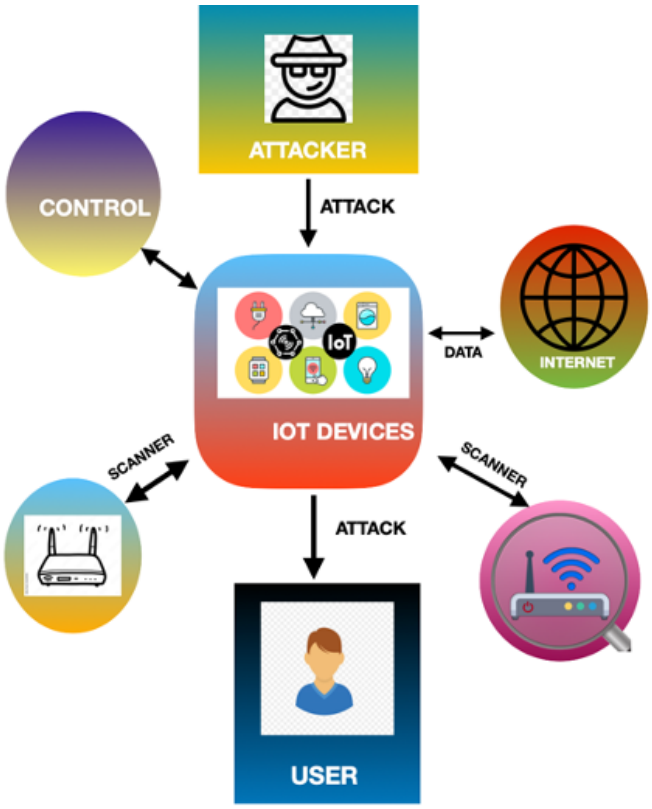


Fig. 4. Internet of Things Vulnerabilities

Link to this paper's aims: The breadth of attack techniques and persistence of exploitable conditions highlight a central need: a larger, better-prepared cybersecurity workforce that understands foundational security principles early and applies them consistently. This paper's proposed Grades 7–12 curriculum addresses that need by cultivating baseline knowledge and hands-on competencies aligned to industry frameworks.

B. Cause of Cyberattacks

The frequency and variety of cyberattacks continue to rise across academic, industrial, and media reporting, prompting questions about the extent to which modern society may be disrupted by cyber operations [20], [21]. Causes are multi-factorial, spanning human behavior, organizational practices, technology complexity, and systemic constraints.

Human and organizational factors. A lack of knowledge in securing technological components, from software development to network administration, creates exploitable gaps when teams are unfamiliar with the environments they protect. Insufficient awareness in security operations can increase false positives/negatives, diverting attention from genuine incidents. At the end-user level, low cybersecurity awareness remains a persistent risk; for example, many employees underestimate the likelihood of mobile compromise from suspicious links [22]. Broader education system gaps contribute to pipeline shortages: middle and high school students often lack exposure to cybersecurity and digital forensics as viable and meaningful career pathways.

Technology and market dynamics. The rapid pace of technological change demands continuous learning and retraining. Concurrently, high demand for qualified practitioners outstrips supply, and some students gravitate toward adjacent fields (software engineering, data science), leaving cyber roles unfilled. Within the taxonomy of cyber-dependent crimes, offenses typically fall into two categories: illicit intrusions into computer networks (e.g., hacking) and disruption/downgrading of computer functionality and network space (e.g., malware, DDoS) [23].

Common reasons exploited by attackers. System accessibility and heterogeneity allow multiple intrusion vectors, including covert logic bombs, keyloggers capturing credentials, and attempts to defeat biometric controls [24], [25]. High data storage efficiency enables compact retention and rapid retrieval/modification, facilitating mass exfiltration or tampering [23], [25]. Code complexity in modern operating systems and applications yields inevitable defects that adversaries weaponize [25]. Human negligence, from misconfigurations to lax access controls, provides low-effort paths to compromise [26]. Finally, evidence loss complicates investigations, especially where data destruction, jurisdictional boundaries, or privacy constraints impede forensic procedures [23], [25], [27].

Impacts. Consequences range from large-scale fraud and theft by organized groups to geopolitical frictions as nations attribute interference in domestic affairs. At the individual level, victims may experience psychological distress or mistrust of technology. The dark web's growth amplifies illicit marketplaces and services, further eroding public confidence in using digital channels for sensitive activities [28], [29].

Link to this paper's aims: These causes and impacts point to three levers, education, regulation, and policy, that must operate in concert. The curriculum proposed in this paper targets the education lever by instilling foundational security concepts and habits early, building a pipeline that can reduce preventable errors and strengthen organizational resilience [23], [28].

C. Scarcity of cybersecurity talent in education, public, and private sectors

Technology is evolving rapidly and becoming more complex, prompting organizations across sectors to deploy new defense capabilities and to recruit professionals who can operate and administer them. Employers report persistent difficulty meeting workforce needs, resulting in overloaded roles and deferred security tasks (*Figure 5*). In practice, some organizations fill positions with candidates from non-security backgrounds who hold selected certifications but lack foundational IT knowledge, limiting effectiveness in prevention and response [30].



Fig. 5. Cybersecurity skill shortage crisis

Public-sector challenges mirror and magnify these pressures. The Department of Defense Science Board highlighted in 2017 that the U.S. military's extensive dependence on information technology increases the attack surface and that major powers can put critical infrastructure at risk over the next decade. The Board concluded that adversaries' offensive capabilities may outpace U.S. defenses and resilience measures in the near term. Workforce projections underscore the scale of the problem: the Global Information Security Workforce Study identified a significant gap, estimating a shortage of 1.8 million workers by 2022; concurrent analytics have shown hundreds of thousands of open roles at various points in time. Former Federal CIO Tony Scott estimated more than 10,000 federal cyber openings in 2015, emphasizing the need for pipeline programs that recruit, train, and retain talent in government cybersecurity [31]. Recent CyberSeek data indicates a severe cybersecurity personnel shortage in the USA with ~514,359 unfilled job openings and only 74% of the needed workforce are currently in place [46].

Education and industry face parallel obstacles in hiring and instruction. Employers struggle to align job descriptions (degree + certifications + 3+ years' experience) with applicant readiness, and colleges/universities face a scarcity of qualified instructors for cybersecurity courses. These filters unintentionally exclude promising candidates lacking formal degrees or hands-on experience, leaving essential positions unfilled and constraining capacity in both practice and teaching.

Link to this paper's aims. The proposed Grades 7–12 curriculum seeks to expand the upstream talent pool, improving preparedness for entry-level pathways, foundational certifications, and further study, thereby easing hiring bottlenecks and creating clearer on-ramps into public and private sector roles.

D. Scarcity of digital forensics experts

Demand for digital forensics is rising across corporations and law enforcement. Market analyses indicate strong growth in the North American digital forensics sector, from USD 1.80B (2021) toward USD 4B by 2028, with hardware tools playing a central role in acquisition, storage, and analysis as shown in *Figure 6* [32]. Simultaneously, organizations reported a steep increase in weekly cyberattacks in 2021, each incident requiring prompt investigation and coordinated response. Law enforcement agencies confront sustained caseload expansion while contending with budget constraints and staff shortages, leading non-technical reviewers to prepare cases and inadvertently introducing errors, backlogs, and bottlenecks [32].

Time sensitivity compounds risk: approximately 45% of crimes occur between 7 p.m. and 7 a.m., outside standard working hours, stretching investigation teams and response windows [33], [34]. Agencies sometimes rely on forensic lab subcontractors to accelerate case preparation overnight, but at substantial cost (e.g., ~USD 80,000 per subcontractor annually), with higher totals when multiple staff and benefits are involved. For corporations unable to fill forensic roles, the principal cost is delayed incident response: ransomware investigations and integrations with SOAR/SIEM tooling can be slowed by staffing gaps and manual processes, raising the likelihood of data loss, extortion pressure, and regulatory exposure.

Link to this paper's aims: By introducing foundational forensics concepts and hands-on labs within the Grades 7–12 pathway, the curriculum builds early competence in evidence handling, acquisition, and analysis, preparing students for further specialization and helping alleviate pressure on investigative teams.

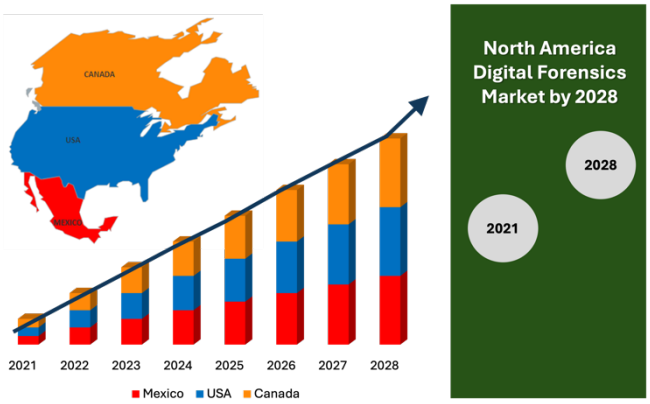


Fig. 6. *Digital Forensics Experts Market in the US*

E. Role of community colleges and universities in cybersecurity education

Given the substantial cybersecurity talent shortfall, community colleges and universities play a pivotal role in expanding national capacity. Institutions can bridge disparities by offering targeted programs (e.g., Bachelor's in Cybersecurity, IT majors with cybersecurity concentrations, CS minors in security) and by establishing graduate tracks in IT/security to develop the instructor pipeline needed across schools, colleges, and universities.

Many post-secondary institutions participate in the National Centers of Academic Excellence in Cybersecurity (CAE-C) program, which establishes nationally recognized curricular standards through defined Knowledge Units (KUs). CAE-designated programs in Cyber Defense, Cyber Operations, and Research align coursework with workforce needs and national security priorities, while promoting continuous program assessment and improvement. In parallel, institutions increasingly map curricula to the NIST NICE Cybersecurity Workforce Framework, which defines work roles along with associated tasks, knowledge, and skills, and to the DoD Cyber Workforce Framework (DCWF) for defense-related roles. These frameworks provide a common language linking education, certification, and employment, ensuring graduates are prepared for clearly defined cybersecurity career pathways and emerging specialty areas.

Community colleges provide accessible entry points through associate degrees, certificates, stackable credentials, and articulation agreements with CAE-designated universities, enabling seamless transfer into bachelor's programs. These institutions are particularly important in broadening participation and expanding regional workforce capacity. Academic advisors play a critical role at enrollment, guiding undecided students toward cybersecurity's societal importance and strong career trajectories using structured workforce role definitions.

Collaboration between education and industry, through internships, apprenticeships, cyber labs, and capstone partnerships, ensures curricula reflect real-world enterprise environments and evolving threat landscapes. Such partnerships also strengthen feedback loops between employers and educators, supporting agile curriculum updates in response to changing threat vectors and technologies.

Link to this paper's aims: The proposed Grades 7–12 program complements higher-education pathways by introducing foundational competencies aligned with CAE Knowledge Units and national workforce frameworks. Early exposure supports readiness for dual-credit, certificate, or associate programs at community colleges and facilitates smoother progression into university degrees and entry-level cybersecurity roles.

F. Related Work on Cybersecurity Education and Pedagogical Approaches

Early STEM interventions. Research shows that early exposure to STEM and computing increases persistence and later participation in advanced coursework; middle-school introductions to computing concepts correlate with higher enrollment in high school pathways [35], [36]. National initiatives, GenCyber (NSA) and CyberPatriot (AFA), demonstrate measurable gains in awareness and engagement. However, their episodic formats (camps, competitions) limit continuity and integration into credit-bearing curricula, reinforcing the value of structured, multi-year programs embedded in school systems [37], [38].

Competency frameworks (NICE). The NICE Cybersecurity Workforce Framework (NIST SP 800-181 Rev. 1) organizes work roles, tasks, knowledge, and skills across security domains

[14]. Aligning secondary course objectives and assessments with NICE K/S/A, spanning *Securely Provision*, *Operate and Maintain*, *Protect and Defend*, and *Investigate*, supports career exploration and scaffolds preparation for foundational certifications (e.g., A+, Network+, Security+) and entry-level roles.

Pedagogical approaches. Evidence-based strategies support engagement and skill acquisition. *Gamification* (e.g., points, badges, team challenges) increases motivation and persistence when tightly aligned to learning outcomes rather than novelty [39]. *Inquiry-based learning* (questioning, hypothesis formation, investigation, data analysis, reflection) cultivates critical thinking essential to cybersecurity problem-solving and incident analysis [40]. *Problem-based labs* situate learners in authentic scenarios, configuring firewall policies, triaging phishing incidents, performing basic forensic acquisitions, bridging theory and practice and improving transfer to workplace tasks [41]. Collectively, these approaches complement the proposed Grades 7–12 progression, from hardware/OS fundamentals through networking, security, and digital forensics, maintaining engagement, developing higher-order thinking, and building practical competence aligned to NICE.

Integrative link: The literature reviewed above substantiates the need for earlier, sustained, and framework-aligned cybersecurity education. It also informs our research methodology (curriculum mapping to NICE; hands-on labs; pilot surveys) and provides a basis for interpreting findings regarding student awareness, interest, and preferences for experiential learning. By connecting workforce shortages, attack trends, and pedagogy, Section 2 motivates the curriculum design decisions evaluated in the remainder of this paper

III. Research Methodology and Student Survey

A. Research Methodology

The development of the proposed Grades 7–12 cybersecurity curriculum followed a structured process combining literature synthesis, expert consultation, and alignment with established competency frameworks. Specifically:

- Literature Review & Needs Analysis: A comprehensive review of academic and industry sources was conducted to identify workforce gaps, educational deficiencies, and best practices in cybersecurity education [1]–[4], [27]–[34].
- Expert Panels: Input was solicited from five cybersecurity professionals and two K–12 educators to ensure curriculum relevance and pedagogical appropriateness.
- Framework Adaptation: The curriculum was mapped to the NICE Cybersecurity Workforce Framework (NIST SP 800-181 Rev. 1) [30], ensuring alignment with industry-recognized knowledge, skills, and abilities.

Pilot Survey: To gauge baseline awareness and interest, a survey was administered to 137 students across four private schools in the Chicagoland area.

- Sample Size & Selection: Participants included students in grades 7–12, selected through voluntary participation after parental consent.
- Data Collection: A 10-question instrument assessed familiarity with cybersecurity concepts, interest in technology, prior participation in competitions, and perceived importance of cybersecurity careers. Responses were collected anonymously via paper forms and aggregated for analysis.
- Ethical Considerations: The study adhered to ethical guidelines for research involving minors. Parental consent was obtained for all participants, and no personally identifiable information was collected. The survey was approved by the participating schools' administrative boards.

This mixed-method approach ensured that the curriculum design was evidence-based, expert-informed, and aligned with national standards, while the pilot survey provided preliminary insights into student awareness and interest

B. Survey of 7th to 12th Grade Students

The previous section on the *present state of cyberattacks and the scarcity of qualified cybersecurity talent* extensively covered four pivotal areas and highlighted pressing issues in the cybersecurity domain while underscoring the urgent need for solutions to bridge the talent gap and fortify digital defenses in the United States. With this in mind, this research manuscript's next step was to determine the understanding and knowledge of Cybersecurity and Digital Forensics areas among 7th to 12th-grade students in middle and high schools.

Toward this goal, four private schools in the greater Chicagoland area were surveyed with a set of 10 questions gauging the extent of exposure and knowledge of 7th to 12th graders. A total of 137 students responded to the survey, and the responses to various questions are displayed in the form of pie charts. *Figure 7* shows the student participation results in the form of percentages for various grades.

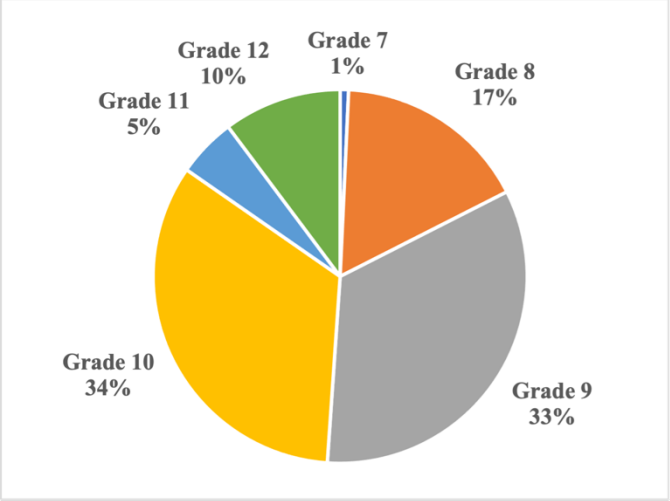


Fig. 7. *Student participation percentage based on grade level*

The highest level of participation in the survey was from 10th, then, 9th, and then 8th graders totalling 84%, while the combined participation of 12th, 11th, and 7th graders was limited to 16%. Examining the survey takers based on gender, at least 50% of survey takers were Female students as shown in *Figure 8*.

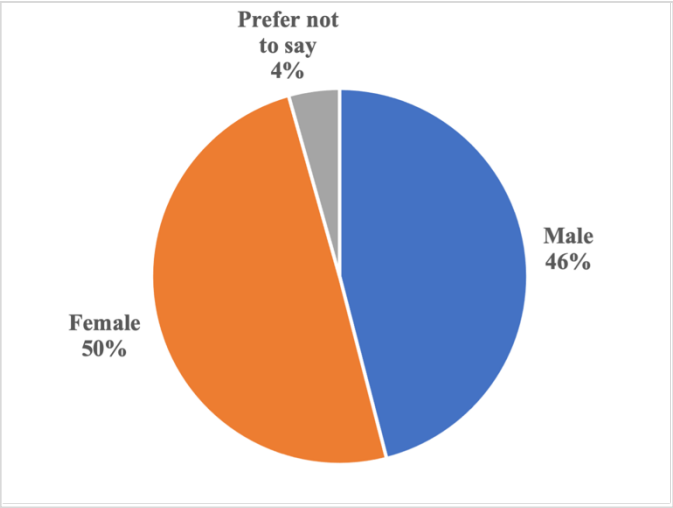


Fig. 8. *Student participation level based on gender*

When asked about their familiarity with the subject of cybersecurity and digital forensics, 77% of the survey takers said they had heard about this area before taking the survey, while 23% of the survey takers had not heard of this area. The results of this question are shown in *Figure 9*. It is noteworthy that despite the relatively low awareness among some participants, the overall interest in cybersecurity appears to be substantial.

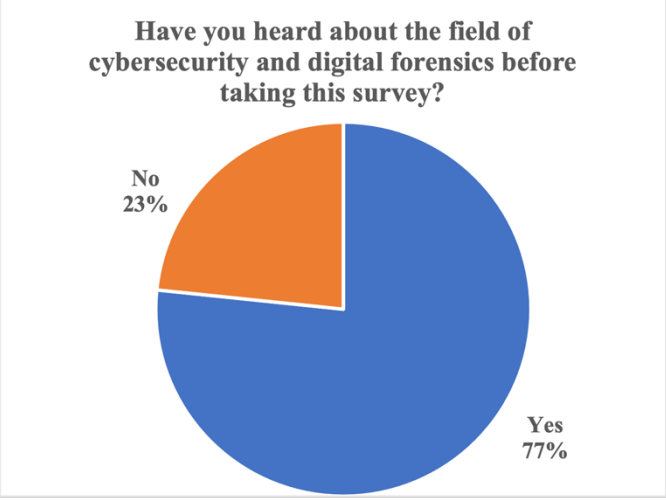


Fig. 9. *Participant's familiarity with cybersecurity and digital forensics*

When asked if they had any general interest in Computer Science and Technology, 50% of survey takers expressed interest, while the remaining 50% said that they were not interested. The results of this question are shown in *Figure 10.* Exploring the reasons behind the lack of interest among some respondents could provide valuable insights into potential areas for improvement in promoting Computer Science and Technology education.

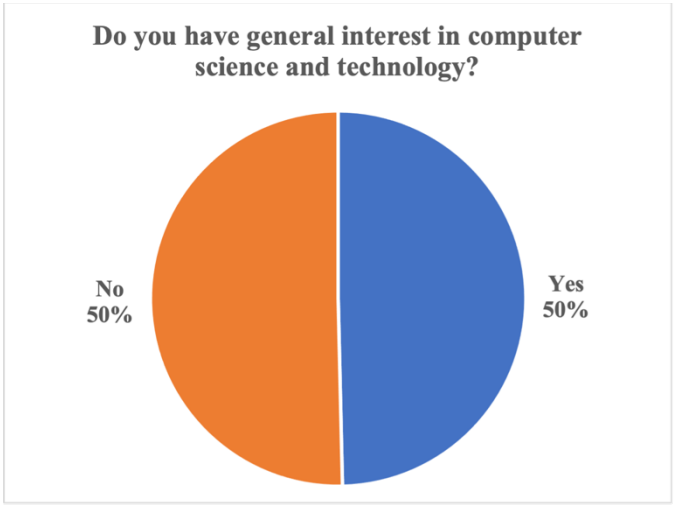


Fig. 10. *Participant's general interest in computer science and technology*

When asked if they had participated in any cybersecurity competition, 93% of the participants said no. The results of this question are shown in *Figure 11*. Considering the increasing importance of cybersecurity skills, further investigation into the factors hindering participation in such competitions might help in designing initiatives to encourage more active engagement.

On the other hand, when asked whether they believe that cybersecurity and digital forensics are important fields for the future, 89% of the participants responded affirmatively. The results of this question are shown in Figure 12. Because the question combined both areas, the responses reflect students' perceptions of the broader cybersecurity/digital forensics domain rather than either field independently. This strong vote of confidence underscores the need for educational programs and awareness campaigns to bridge the gap between general interest and active participation in cybersecurity-related activities.

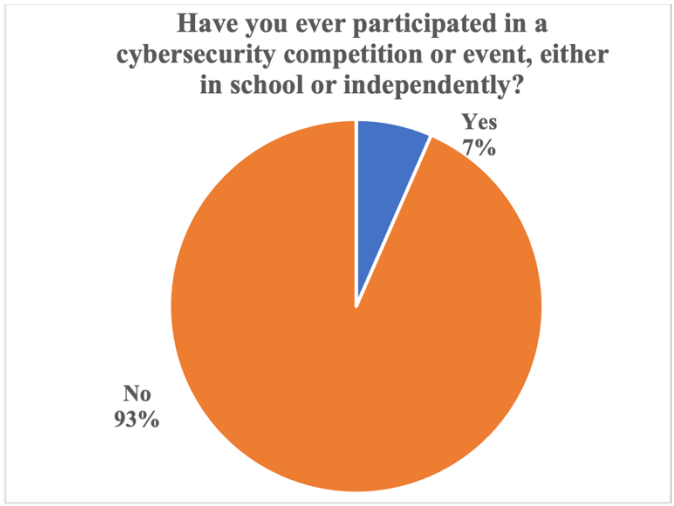


Fig. 11. *Students' response to participation in cybersecurity competition*

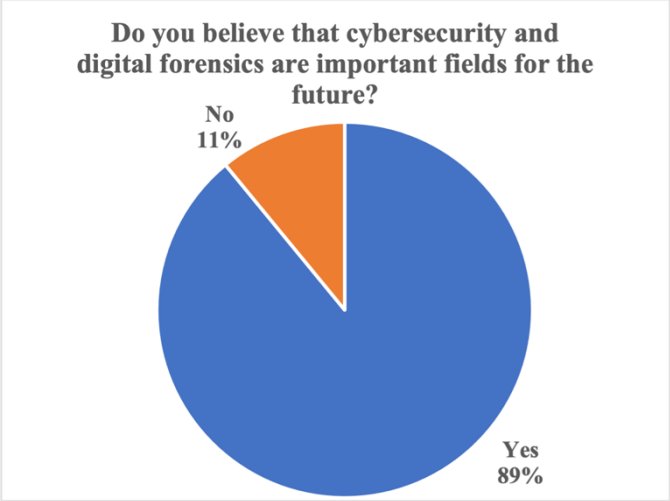


Fig. 12. *Student response to the importance of cybersecurity and digital forensics field.*

Finally, the survey participants were asked what schools should do to prepare students for careers in cybersecurity and digital forensics, a whopping 67% of the participants wanted the schools to offer a cybersecurity curriculum with hands-on labs and workshops. The results of this question are shown in *Figure 13*.

The survey revealed that 93% had not participated in any cybersecurity competition. Despite this, 89% believed that cybersecurity and digital forensics are important fields for the future. In terms of preparation for careers in these fields, 67% of participants advocated for schools to offer a cybersecurity curriculum with hands-on labs and workshops. These findings underscore both the existing interest among students and the need for more educational initiatives in cybersecurity and digital forensics.

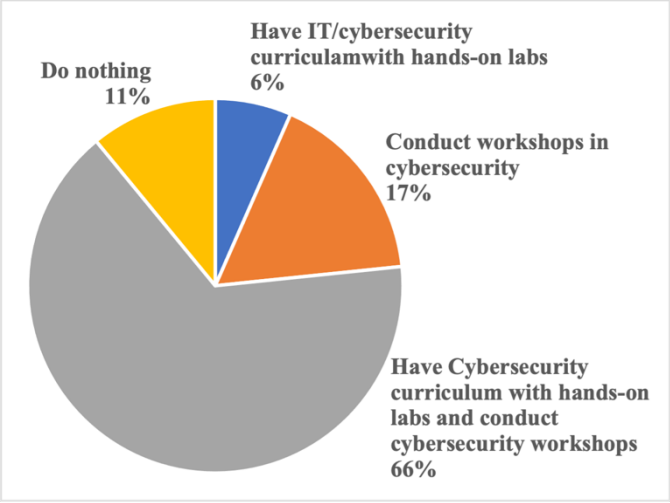


Fig. 13. *Student response to the role of schools in cybersecurity.*

IV. Need for IT/Cybersecurity Programs in Middle and High Schools

The persistent shortage of cybersecurity professionals is closely linked to the limited and inconsistent integration of formal information technology and cybersecurity courses in many middle and high school curricula. While several states provide Career and Technical Education (CTE) pathways that include networking and cybersecurity strands [42], and national organizations such as Teach Cyber, Cyber.org, and the CSEC 2025 curricular guidelines offer structured frameworks and resources [43], [44], [45], implementation and depth vary across districts. Current K–12 frameworks often emphasize general computing literacy but less frequently incorporate sustained instruction in hardware, networking, programming, and security fundamentals. This uneven exposure leaves many graduates unaware of cybersecurity as a discipline and underprepared for advanced study or entry-level roles.

Although technologies evolve rapidly, foundational knowledge remains essential. Early exposure to core concepts, such as operating systems, networking, and secure practices, provides a durable base for adapting to future innovations. Introducing cybersecurity fundamentals in middle school can serve as a gateway to deeper engagement in high school, enabling students to pursue advanced coursework, certifications, and pathways into college programs in cybersecurity and digital forensics.

National initiatives such as GenCyber [37] and CyberPatriot [38] have raised awareness and engagement; however, their episodic structure may limit curricular continuity if not embedded within a sustained program of study. Aligning secondary curricula with recognized standards such as the NICE Workforce Framework for Cybersecurity [14] can further strengthen coherence between K–12 education and workforce competencies.

Building on several years of classroom implementation and iterative refinement in middle and high school settings, this paper proposes a structured Grades 7–12 curriculum aligned with existing state and national guidance while emphasizing hands-on laboratories and competency-based progression. One complementary strategy is the potential development of an Advanced Placement (AP) Cybersecurity course through the College Board, which could further elevate the visibility and rigor of cybersecurity education nationwide and support sustained pathways into cybersecurity careers.

V. Proposed IT/Cybersecurity Curriculum of 7th to 12th Grades

A. Curriculum Structure

The proposed Grades 7–12 curriculum is organized into six progressive modules, each spanning two semesters:

- Grade 7: IT 101 – Computer Hardware Fundamentals
- Grade 8: IT 102 – Computer Operating Systems
- Grade 9: IT 103 – Computer Networking Fundamentals
- Grade 10: IT 104 – Introduction to Network Security-1
- Grade 11: IT 105 – Introduction to Network Security-2
- Grade 12: IT 106 – Computer Forensics and Investigations

Details of the curriculum for Grade 7-12 are provided in **Appendix** at the end of the manuscript.

B. Learning Objectives

Each module includes objectives aligned with NICE KSAs, covering hardware assembly, OS installation, networking protocols, cryptography basics, firewall configuration, intrusion detection, and forensic analysis.

C. Assessment Methods

- Formative Assessments: Weekly quizzes, lab exercises, and troubleshooting tasks.
- Summative Assessments: End-of-semester practical exams, project-based evaluations (e.g., configuring a secure network, conducting a forensic investigation).
- Optional Certifications: Students are encouraged to attempt CompTIA A+, Network+, and Security+ after completing relevant modules.

D. Pilot Survey Findings

Pilot Survey Findings: The survey of 137 students revealed:

- *Awareness:* 77% had heard of cybersecurity concepts.
- *Perceived Importance:* 89% agreed cybersecurity and digital forensics are critical for the future.
- *Interest in Hands-On Learning:* 67% wanted schools to offer labs and workshops.
- *Engagement Gap:* 93% had not participated in any cybersecurity competition, indicating a need for structured programs.

E. Implementation Challenges

- *Teacher Training*: Few instructors possess both technical expertise and pedagogical skills; graduate programs and industry partnerships are essential.
- *Infrastructure Needs*: Schools require computer labs with networking equipment, virtualization tools, and forensic software.
- *Policy Alignment:* Curriculum integration must align with state education standards and College Board requirements for AP Cybersecurity.

F. Comparison to Existing Programs

Unlike CyberPatriot and GenCyber, which focus on short-term competitions and camps, this curriculum offers a multi-year, formal pathway embedded in school systems. It emphasizes progressive skill development, hands-on labs, and alignment with NICE competencies, preparing students for certifications and careers.

VI. Benefits of the Proposed Curriculum

In today's rapidly evolving digital landscape, a curriculum focused on cybersecurity and digital forensics is essential and aligns with national and international standards aimed at bridging the workforce gap. The proposed Grades 7–12 curriculum is intentionally designed as a scaffolded progression, beginning with foundational hardware and software concepts and advancing toward networking, cybersecurity principles, and digital forensics to reflect cognitive development and skill maturity across grade levels. Course selection and sequencing are guided by pedagogical scaffolding principles, workforce competency requirements, and alignment with established frameworks such as the Cybersecurity Education Consortium (CSEC) standards, the K–12 Cybersecurity Learning Standards, and the NIST NICE Cybersecurity Workforce Framework (SP 800-181 Rev. 1). This alignment ensures that students acquire competencies recognized across education and industry, positioning them for success in both academic and professional pathways.

The curriculum incorporates core concepts emphasized by CSEC and K–12 Cybersecurity Standards, including secure computing practices, risk management, and ethical considerations. By integrating these standards into age-appropriate modules, students gain exposure to cryptography, network defense, and digital evidence handling, which are critical for understanding modern security challenges. Mapping to the NICE Framework further strengthens the curriculum by aligning course objectives with foundational work roles and knowledge areas such as Securely Provision, Protect and Defend, and Investigate. This competency-based structure ensures that each course builds upon prior knowledge, reinforcing technical depth, problem-solving ability, and applied skills relevant to real-world cybersecurity tasks while supporting early career exploration.

Completion of the curriculum also prepares students for widely recognized entry-level certifications, including CompTIA IT Fundamentals (ITF+), A+, Network+, and Security+. The program introduces hardware and operating system fundamentals, networking concepts, and basic security principles, which are prerequisites for A+ and Network+. Modules on threat identification, risk mitigation, and cryptography provide the conceptual grounding for Security+. Hands-on labs and simulations reinforce practical skills, bridging theory and application, which is critical for success in performance-based certification exams. By the time students graduate from high school, they will have the foundational knowledge and applied experience necessary to pursue advanced coursework at community colleges and universities or to begin certification pathways that lead directly to entry-level roles.

Beyond technical preparation, the curriculum emphasizes holistic skill development through inquiry-based learning, gamified challenges, and problem-based labs. These pedagogical strategies foster critical thinking, adaptability, and problem-solving skills, enabling students to analyze vulnerabilities, implement secure configurations, and conduct basic forensic investigations while adhering to legal and ethical guidelines. The curriculum is presented as an adaptable, standards-aligned framework that schools may tailor to their infrastructure, scheduling models, staffing expertise, and state or district requirements, rather than as isolated standalone syllabi, ensuring maximum flexibility for diverse educational settings worldwide. This combination of theoretical knowledge, practical experience, structured progression, and standards alignment ensures that graduates of the program are well-positioned to meet the growing demand for cybersecurity professionals and to contribute meaningfully to securing digital infrastructure.

By embedding cybersecurity education into middle and high school programs and aligning it with recognized standards and frameworks, this curriculum addresses a key root cause of the workforce shortage: lack of early exposure and structured learning. Students completing this program will enter higher education or the workforce with foundational competencies, practical experience, and a clear trajectory toward industry-recognized credentials, thereby strengthening the cybersecurity talent pipeline and enhancing national resilience.

VII. Discussion and Conclusions

A. Discussion

The scarcity of cybersecurity and digital forensics professionals remains a critical and persistent challenge for educational institutions, industry stakeholders, and governments worldwide. This shortage is driven by multiple factors, including gaps in formal education, limited awareness of career opportunities, and the rapid evolution of technology that outpaces traditional academic curricula. Although educational programs and certifications exist, many lack sufficient hands-on training and practical exposure, which are essential for preparing students to address real-world cybersecurity threats. Furthermore, the shortage of qualified instructors exacerbates these challenges, creating bottlenecks in scaling cybersecurity education.

Awareness is another significant barrier. A large segment of students and professionals remains unaware of the importance and viability of careers in cybersecurity and digital forensics. Without targeted outreach and early exposure, these fields fail to attract the talent needed to meet growing demand. The dynamic nature of cybersecurity compounds the problem, as new technologies, vulnerabilities, and attack vectors emerge continuously, requiring professionals to remain adaptable and engaged in lifelong learning.

Collaboration among academia, industry, and government is vital to overcoming these challenges. Partnerships that foster knowledge sharing, mentorship programs, and apprenticeships can accelerate skill development and create pathways for students to transition into the workforce. While technological solutions such as automation and artificial intelligence offer promising support, human expertise remains indispensable for interpreting complex scenarios, making ethical decisions, and responding to sophisticated threats. Ultimately, the imbalance between demand and supply underscores the urgency of implementing systemic solutions, including curriculum innovation, teacher training, and policy alignment.

B. Conclusions

The proposed Grades 7–12 cybersecurity curriculum addresses these challenges by introducing a structured, multi-year learning pathway that begins with foundational IT concepts and progresses to advanced topics such as network security, cryptography, and digital forensics. Each module incorporates hands-on labs, formative and summative assessments, and alignment with the NICE Cybersecurity Workforce Framework to ensure industry relevance and certification readiness. Pilot survey results from 137 students across four schools revealed strong interest in cybersecurity education, with 77% reporting awareness of cybersecurity concepts, 89% recognizing its importance for the future, and 67% advocating for hands-on labs. These findings highlight both the potential impact of early interventions and the current lack of structured exposure.

Compared to existing initiatives such as CyberPatriot and GenCyber, which primarily focus on short-term competitions and camps, the proposed curriculum offers a formal, progressive, and competency-based approach embedded within school systems. By preparing students for certifications like A+, Network+, and Security+, and by fostering critical thinking through experiential learning, this program creates a sustainable pipeline of talent equipped to meet the demands of a rapidly evolving cybersecurity landscape. In conclusion, significant progress in mitigating cyber threats can be achieved through investments in education, heightened awareness, collaborative efforts, and diversity initiatives. These measures collectively contribute to building a robust foundation for a safer and more resilient digital future

VIII. Recommendations and Future Work

This section consolidates practical recommendations for near-term implementation with research directions that support continuous improvement. The dual focus ensures that schools and partners can act immediately (teacher preparation, infrastructure, policy alignment, and scaling) while the research community conducts longitudinal studies to validate impact, refine pedagogy, and anticipate evolving threats.

This section outlines actionable recommendations for key stakeholders, a phased roadmap for scaling the Grades 7–12 cybersecurity curriculum, and strategies for impact evaluation to ensure meaningful contribution toward closing the cybersecurity workforce gap.

1). Actionable Steps for Stakeholders

a) Policymakers and Agencies

- Establish state and national standards for K–12 cybersecurity education aligned with the NICE Cybersecurity Workforce Framework.
- Allocate dedicated funding for teacher professional development, infrastructure upgrades, and curriculum integration.
- Develop policies that incentivize schools to adopt cybersecurity education and support certification pathways for students.
- Issue guidance for safe, ethical, and compliant operation of labs (e.g., use of isolated networks, logging, privacy protection for minors), and streamline IRB-like approvals for school-based studies.

b) Schools and Educational Institutions

- Integrate cybersecurity modules into existing curricula, ensuring alignment with academic standards and career readiness frameworks.
- Invest in laboratory infrastructure, including networking equipment and forensic tools, to facilitate hands-on learning.

- Implement teacher training programs to build instructional capacity and maintain curriculum relevance.
- Offer certification preparation (e.g., Security+, CYSA, or vendor-neutral equivalents for secondary level), dual-enrollment options, and capstone projects with community partners.

c) Industry and Private Sector

- Collaborate with schools to provide mentorship, internships, and real-world project opportunities for students.
- Offer resources such as software licenses, hardware donations, and access to cybersecurity tools.
- Participate in curriculum advisory boards to ensure alignment with evolving workforce needs and emerging technologies.
- Partner to reduce barriers for underserved schools (equipment pools, mobile labs, stipend support, remote mentoring).

2) Roadmap for Scaling the Program

a) Phase 1: Pilot Implementation (Year 1)

- Deploy the curriculum in select schools across diverse regions.
- Collect feedback from students, teachers, and administrators to refine modules and assessments.

b) Phase 2: Regional Expansion (Years 2–3)

- Scale implementation to additional districts, supported by teacher professional development programs.
- Establish regional cybersecurity education hubs for resource sharing and collaboration.

c). Phase 3: Nationwide Adoption (Years 4–5)

- Align curriculum with state and national education standards.
- Secure long-term funding and partnerships with industry and government agencies.
- Integrate certification pathways and dual-enrollment options with higher education institutions.

3) Impact Tracking and Longitudinal Studies

a) To evaluate the effectiveness of the program and inform continuous improvement, longitudinal studies should be conducted over a 3–5 year period, focusing on:

- *Student Engagement:* Participation rates, interest in cybersecurity careers, and enrollment in advanced modules.
- *Skill Development:* Performance in formative and summative assessments, certification attainment, and practical competencies.
- *Career Outcomes:* Tracking graduates' progression into cybersecurity-related higher education programs and workforce roles.
- *Program Sustainability:* Measuring teacher retention, infrastructure utilization, and policy compliance across schools.

b) Data Collection Cadence

- *Short-Term (each term):* Module-level assessments, student/teacher feedback, lab safety audits.
- *Annual:* Certification metrics, pathway transitions, equity analyses, infrastructure utilization, PD effectiveness.
- *3–5 Year Horizon:* Cohort tracking into postsecondary/industry, comparative analyses against non-participating schools, cost–benefit and ROI studies.

c) Reporting and Continuous Improvement

- Produce annual technical reports and dashboards for stakeholders.
- Convene review boards (policymakers, district leaders, industry) to act on findings, updating curricula, PD, and resource allocations.

B. Future Research Directions

Incorporating cybersecurity into middle and high school curricula is imperative in today's digital age, empowering students with the knowledge and skills to protect themselves and contribute to a more secure online environment. Areas for future research and development in cybersecurity education for middle and high schools include, but are not limited to:

- *Emerging Threats and Technologies***:** Ongoing exploration and understanding of emerging cybersecurity threats and technologies to ensure that educational content remains relevant and up to date.
- *Penetration Testing:* Further investigation into the implementation of penetration testing concepts, providing students with practical skills in assessing and strengthening digital security measures.
- *Digital Citizenship:* Exploration of digital citizenship, emphasizing responsible and ethical behavior in the digital realm, fostering a sense of accountability and awareness among students.
- *Integration with Existing Subjects*: Research on seamlessly integrating cybersecurity concepts into established subjects such as computer science, mathematics, and ethics. Examining how these concepts can enhance overall learning and skill development.

These areas represent avenues for future work, contributing to the ongoing enhancement of cybersecurity education in middle and high schools, and ensuring that students are well-prepared to navigate the evolving challenges of the digital landscape.

C. Study Limitations and Disclaimer:

The data in this study were gathered solely from private educational institutions. As a result, the findings may reflect characteristics that are more common in private school settings, such as greater flexibility in curriculum design, faster administrative approval for new technologies, and differences in student demographics. These conditions are not always present in public schools.

For this reason, readers should be cautious about applying these results directly to public or vocational-technical environments, where stricter administrative procedures and standardized curriculum requirements may influence both implementation and outcomes. This study focuses specifically on private institutions, and additional research is needed to evaluate how well the proposed framework would perform in public school settings.

APPENDIX

In this Appendix detailed proposed curriculum for 7th to 12th grades is given in the next 6 columns of this manuscript.

Grade: 7 Proposed IT/Cybersecurity Curriculum

Course Number and Name: IT 101 Computer Hardware Fundamentals.
Course Description: This course teaches the foundations of computer hardware, with an emphasis on helping students understand and learn computer hardware skills with hands-on labs. Once students complete the course, they will be able to identify computer hardware components, understand the function of each part, and install the parts to build a computer.
Course Objective(s)

Students will be able to:

1. Identify all the computer hardware components.
2. Understand the different form factors of computers.
3. Learn the function of all the hardware components.
4. Install the major hardware components.
5. Troubleshoot the basic problems related to computer hardware.

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact Information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu
Textbook: Textbook: A+ Guide to Hardware by Jean Andrews 9th Edition 2017 ISBN-13:9781305266452

Grade 7 Course Outline: IT 101 Computer Hardware Fundamentals

Weeks	Course Topics
1-6	a) Introduction to Computers b) System Architecture of Desktops and Laptops
7-12	c) Input/output devices and physical ports d) Motherboards
13-18	e) Processors f) Memory
19-24	g) Power system h) Storage devices and partitions
25-30	i) Memory j) Network and Graphic cards
31-36	k) Firmware interface and Drivers l) Basic hardware Troubleshooting

Grade: 8 Proposed IT/Cybersecurity Curriculum

Course Number and Name: IT 102 Computer Operating Systems.
Course Description: This course teaches the foundations of computer operating system fundamentals, with an emphasis on helping students understand and learn computer operating systems with hands-on labs. Once students complete the course, they will be able to install, configure and troubleshoot operating systems.
Course Objective(s)

Students will be able to:

1. Understand the function of an Operating System.
2. Understand different Operating systems and versions.
3. Install Operating System.
4. Understand the file system of Windows and Linux OS.
5. Troubleshoot the basic problems related to OS.

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu
Textbook: A+ Guide to Operating Systems and Security by Jean Andrews 10th Edition, 2020 ISBN 78-0357108505

Grade 8 Course Outline: IT 102 Computer Operating Systems

Weeks	Course Topics
	Windows Operating Systems
1-3	a) Introduction to desktop operating systems b) Types and versions
4-7	c) Installing Microsoft operating system d) Basics of Filesystem
8-11	e) Embedded Features of Microsoft OS f) Introduction to Windows applications g) Command Line Interface and PowerShell commands
12-15	h) Windows Network Operating systems i) Introduction to Windows IIS server j) Windows O/S Troubleshooting
	Linux Operating System
16-18	k) Introduction to the Linux Operating System l) Types and versions of Linux Operating systems
19-24	m) Linux Kernel n) Intro to Kali Linux
25-36	o) List of Commands and its functions p) Command Shell q) Embedded features of Kali Linux

Grade: 9 Proposed IT/Cybersecurity Curriculum

Course Number and Name: IT 103 Computer Networking Fundamentals
Course Description: This course teaches the foundations of computer Networking Fundamentals, with an emphasis on helping students understand and learn computer networking skills with hands-on labs. Once students complete the course, they will be able to install networking devices, create local area networks, and troubleshoot the most common networking issues.
Course Objective(s)

Students will be able to:

1. Understand Networking basics.
2. Identify computer network topologies.
3. Understand the OSI layers and their functions.
4. Understand the basics of application layer and transport layer protocols.
5. Understand the basics of IPv4 and IPv6 addressing.
6. Understand the basic function of Networking devices and wireless networks.

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu\
Textbook: A+ Guide to Operating Systems and Security by Jean Andrews 10th Edition, 2020 ISBN 78-0357108505 - Cengage

Grade 9 Course Outline: IT 103 Computer Networking Fundamentals

Weeks	Course Topics

1-6	a) b)	Introduction to computer networking Networking Architecture
7-12	c) d)	Client-Server architecture OSI Model
13-18	e) f)	Application layer and its protocols Basics of TCP, UDP and port numbers
19-24	g) h)	IPv4 and IPv6 Common Networking Devices
25-30	i) j)	Routers Switches and MAC address basics
31-36	k) l)	Wireless Network fundamentals Wireless Access points

Grade: 10 Proposed IT/Cybersecurity Curriculum

Course Name and Number: IT 104 Introduction to Network Security-1
Course Description: This course provides a detailed review of the network security field, including essential terminology, the history of the discipline, and practical techniques to manage the implementation of network security solutions. After an overview of information, network, and web security, students will explore network perimeter defence technologies and methods, including access controls, firewalls, and VPNs.
Course Objective(s)

Students will be able to:

1. Demonstrate a basic understanding of network security with hands-on labs.
2. Cryptography basics
3. Firewall and VPN basics

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu
Textbook: Guide to Network Security Fundamentals by Mark Ciampa 6th Edition, 2018, ISBN: 9780357686256 – Cengage

Grade 10 Course Outline: IT 104 Introduction to Network Security-1

Weeks		Course Topics
1-6	a)	Introduction to security
7-12		Malware and Social Engineering Attacks
13-18	b)	Cryptography
19-24	c)	Advanced Cryptography and PKI
25-30 31-36	d)	Networking and Server Attacks

Grade: 11 Proposed IT/Cybersecurity Curriculum

Course Name and Number: IT 105 Introduction to Network Security-2
Course Description: This course provides students to explore network perimeter defense technologies and methods, including access controls, firewalls, and intrusion detection systems, as well as applied cryptography in public key infrastructure, wireless security, and application security. Assessment of network security, professional careers in the field, and contingency planning are also covered.
Course Objective(s)

Students will be able to:

1. Demonstrate a basic understanding of IDS and IPS
2. Basics of wireless networks and vulnerabilities
3. Basics of network Monitoring and logging
4. Basics of digital forensics

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu
Textbook: Guide to Network Security Fundamentals by Mark Ciampa 6th Edition, 2018 ISBN: 9780357686256 – Cengage

Grade 11 Course Outline: IT 105 Introduction to Network Security-2

Weeks		Course Topics
1-6	a)	Network Security Devices, Design and Technology
7-12	b)	Administering a secure network
13-18	c)	Wireless Network Security
19-24	d)	Client Security
25-30	e)	Physical Security and Application Security
31-36	f)	Identity and Access Management

Grade: 12 Proposed IT/Cybersecurity Curriculum

Course Name and Number: IT 106 Computer Forensics and Investigations
Course Description: This course focuses on the use of the most popular forensics tools and provides specific guidance on dealing with civil and criminal matters relating to the law and technology. Includes discussions on how to manage a digital forensics operation in today's business environment.
Course Objective(s)

Students will be able to:

1. Understand the basics of the digital forensic profession.
2. Learn about the most popular digital forensic tools.
3. Understand the usage of the tools.
4. Learn the steps taken to perform digital forensics investigations.
5. Do data acquisition and report writing.
6. Learn and understand the basics of mobile devices and social media investigation.

Course Length: 2 Semesters
Instructor: John Doe
Class Location: Morgan Hall Room 234
Office Hours: M, W, F 11:30 to 1:30
Contact information: Room No. 23, Burgess Hall, Phone: x2955, email: JohnD@goodschool.edu
Textbook: Guide to Computer Forensics and Investigations 7th Edition by Bill Nelson, Amelia Phillips, Christopher Steuart, Copyright 2025 Available February 2024-cengage

Grade 12 Course Outline: IT 106 Computer Forensics and Investigations

Weeks		Course Topics
1-6	a) b)	Understanding the digital forensics profession and investigations Report Writing and Testimony for digital investigations
7-12	c)	Investigators Laboratory and digital Forensics tools
13-18	d)	Data Acquisition
19-24	e) f)	Processing and incident scenes Working with MS files Systems and Windows registry
25-30	g)	Recovering graphics files and forensics analysis
31-36	h) i)	Mobile device forensics Email and social media investigations.

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